



NITI Aayog

DRAFT



POLICY AT A GLANCE

Ensuring financial sustainability of public bus services

Urban Sector in India

DRAFT

Policy at a Glance

Ensuring financial sustainability of public bus services

Urban Sector in India

The '**Policy-at-a-Glance**' provides users with curated information for first-level analysis and inter-state comparisons of policy. The document brings together all topic-relevant national and state-level acts, policies, and schemes in one place. Further, it identifies for a reader what policy features are covered in which document in a simple table.

The first section of this document sets out some '*Context and Background*' on the topic; the subsequent sections explain the '*Policy Features*' that have been compared and include a '*Glossary*' of technical terms used; the last section is the '*Table of Insights*' that marks out what '*Policy Feature*' has been covered (and on which page) in each of the topic-relevant documents.

Suggested Citation:

NITI Aayog (2023). Ensuring financial sustainability of public bus services. NITI Aayog <Policy-at-a-Glance> Series – Urban Sector in India, 2023-Urban-PAG-05, <link>

Copyright © 2023, NITI Aayog, Government of India.

All rights reserved. This draft document and its contents are internal to NITI Aayog. They cannot be reproduced in any manner either in full or parts, thereof. The draft document is subject to further refinements.

All information, ideas, views, opinions, estimates, suggestions, and recommendations (hereinafter 'content') in this draft "portal resource" should neither be understood as professional advice in any manner nor interpreted as policies, objectives, opinions, or suggestions of NITI Aayog. Readers are advised to use their discretion and seek professional advice before taking any action or decision based on these contents. The contents in this portal have been obtained or derived from reliable sources as far as possible. But, NITI Aayog neither represents this information to be accurate nor assumes any liability for any loss, or damages, caused due to any reason whatsoever, towards its usage by a third party. Readers of this document are made aware that the document may be subject to revisions. Any suggestions/input may please be sent to nitiforstates@gov.in.

www.niti.gov.in

1. Context and Background

City bus services in India generate an aggregate revenue that meets only half of the total expenditure incurred in providing these services. This concern is exacerbated by the fact that city bus services are arguably considerably underprovided in India. Against the benchmark of 40 to 60 buses per lakh population (CEPT University, 2013), Indian cities have deployed public buses well below the set norm. The gap in availability versus demand for bus services is substantially high in small urban cities and towns, as organised bus services are available in only 2.5% of the total urban settlements (only 198 cities out of 7,935 cities; as per Census 2011), compared with 98.1% in million-plus cities (i.e., cities with population of more than ten lakhs). This under-provision of city bus services is exacerbated by the requirement to reduce non-viable services to curb losses. The reduction in services shrinks the coverage and availability of services, thereby increasing dependence on private or informal modes of transport.

Developing viable bus operations require focused interventions, based on internal efficiencies of operations, cost rationalisation, fare revision, performance monitoring and incentivisation. Apart from this, there is a need for a holistic policy to address a clear demarcation of roles and responsibilities of local, state and national governments as well as a need to strengthen capacities of transit agencies to plan, operate and manage the services efficiently to enable development and operations of adequate and quality urban bus-transport systems. The institutional fragmentation in service provision and planning can be addressed through a robust planning framework and an outcome-based approach adopted by each of the service providers for conceptualisation of an integrated view on the mobility requirement of city regions.

The primary objective of this document is to evaluate and compare the key features of various mobility-related policies, plans and frameworks adopted by numerous Indian states. By synthesising information on the urban transport financing frameworks, responsibility frameworks of transport-governing agencies, EV policies and more, this study aims to offer valuable insights into the various approaches and priorities of the states related to financing the sustenance of city bus services.

The cases presented in this document are meant to be illustrative and are not exhaustive. The absence of a city or state in the list is not indicative of the absence of policy in that city or state.

2. Policy Features

This section identifies a common set of policy features that are covered in the documents referred to in Section 3 (Table of Insights).

Financing framework for urban transport

State governments have increasingly tried to enable a dedicated fund to meet the capital requirement for developing the urban-transport infrastructure and during the initial period of operationalising the projects in their cities. The National Urban Transport Policy (NUTP), 2006, recommends setting up dedicated funds in the form of Urban transport funds (UTFs) along with Urban Metropolitan Transport Authorities (UMTAs) in more than a million cities to facilitate coordinated planning and implementation of urban-transport programmes and manage integrated urban transport systems. Currently, Urban Transport Funds are set up through government orders at city level through state government general resolutions or through the provisions in the Urban Metropolitan Authority Transport Authority Act. For example, a state-level government resolution was passed for the formulation of the UMTA in Surat in 2008.

Most state governments and local bodies do not have the requisite financial strength to fund urban transport. As a result, mobilising funds is a formidable challenge, which many Indian states/cities must grapple with. The MoUD guidelines suggest many innovative funding sources that can be tapped, in addition to conventional transport-funding sources. A detailed mechanism for collection of such funds is also provided. Four main streams of funds have been identified, such as collection from (i) public authorities (budgetary allocations from Central and state governments), (ii) direct beneficiaries (additional

user charges), (iii) indirect beneficiaries (betterment levies, land-based fiscal tools, development charges, motor vehicle tax, cess on fuel tax) and (iv) other sources (international funding, borrowings). Provisions are made for operational subsidies for funding of urban transport, including city bus services. For example, the government of Gujarat has a viability gap funding scheme for financing city bus operations to support transport authorities and Urban Local Bodies (ULBs). It provides ULBs with a partial funding support of 50%, or INR 12.5 per km, towards operating expenditure of every new city bus in a Public-Private Partnership (PPP) mode for seven years. Similarly, Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME II) scheme was launched by the Government of India to give a boost to development of Electric Vehicles. The inclusion of capital subsidy in FAME II scheme enabled the sanctioning of 5,595 e-buses all over India across multiple million-plus cities. During the initial phase of the initiative, the Department of Heavy Industries provided State Transport Undertakings (STUs) in nine cities, with a financial support of INR 20,000 per kilowatt-hour to acquire 465 buses. Additional case studies with alternative financing strategies are mentioned below.

Brihanmumbai Electric Supply and Transport (BEST)¹

The BEST suffered more in the past decade as compared to other operators as it had to stop utilising the profit from its electricity distribution wing for the loss-making transport wing as per the provisions under the Electricity Act, 2003. The key sources of revenue for BEST are from passenger fares, bus hires and grants. However, due to low fares and concessions set for maintaining affordability for various categories of commuters, the fare revenue for the operator is not adequate to meet operating costs. The operator has been harnessing other sources of revenue such as advertising and real estate rentals to ramp up its revenues and diversify its sources.

BEST also harnesses non-fare-based revenue sources such as advertising on bus hoardings, in bus televisions, bus stops and other real estate owned by BEST. A fleet of 3,500 buses (three outside panels, and two TVs inside), 35 hoarding sites, 25 bus depots, 47 bus stations, 116 bus chowkies, 3,236 sheltered bus stops and 3,172 pole-bus stops (as of 2017) provides advertising revenue to the operator (Deodhar, 2017).

Apart from collecting advertising revenue, BEST has also monetised the real estate potential of BEST depots and bus stations by leasing the land on a PPP basis for residential or commercial use in lieu of development of bus facilities as well as payment of development premium and lease rent. BEST has been able to collect upfront premium and lease rent across plot areas of 55.6 acres with lease premium of INR 390 crore through acquiring concessionaires for real estate development, station development and bus depots and station development for residential and commercial use in areas of Kurla, Mahim, JVPD etc. Currently, eight sites have been redeveloped under this model, while at least 27 other sites can potentially serve as tremendous source of revenue generation. BEST intends on redeveloping some of these locations (CRISIL study findings, 2022).

Traffic and Transit Management Centers of Bengaluru Metropolitan Transport Corporation (BMTC)²

In 2013, BMTC was the only Public Transport Corporation to serve more than 4.8 million commuters every day within the city of Bangalore. The organisation comprised of a fleet of over 6,600 buses covering an area encompassing a radius of 40 km from the city centre, but it still faced substantial financial losses. In 2017, however, it became one of the very few city bus undertakings to earn a profit by leveraging land holdings in strategic locations throughout the city by the innovative concept of Traffic and Transit Management Centers (TTMCs).

This concept, which combines the development of passenger terminals, with the creation of commercial real estate space, enables BMTC to utilise rental revenue to cross-subsidise the

¹ <https://www.moneylife.in/article/improving-best-revenue-generation-of-mumbais-public-transporter-part2/52034.html>

² https://smarnet.niua.org/sites/default/files/resources/BUS_KARO_2.0-Case_Studies_from_India.pdf

construction cost of the terminal and amenities, and also provide an additional source of revenue for the corporation. The TTMC concept consists of an integrated terminal with adequate facilities and amenities to cater to the requirements of all user groups, a mixed-use development with retail, office space and other commercial activity, to enable people to fulfil all these needs through using bus transport and the provision of park-n-ride facilities to encourage the use of public transport.

BMTC has constructed ten TTMCs since 2009. The initial funding was provided under the Jawaharlal Nehru National Urban Renewal Mission (JnNURM). Subsequently, BMTC was able to recover costs through renting commercial space and invested significant amounts of its own funding in TTMCs. The development of high-quality terminals resulted in significant improvement in the quality of bus travel. BMTC earns roughly INR 15 crore per annum from renting out commercial space. This revenue is expected to increase as commercial spaces are occupied and as rental rates increase due to the strategic location of TTMCs.

Role of UMTAs in city bus services and urban transport financing

A Unified Metropolitan Transport Authority (UMTA) was envisioned in the National Urban Transport Policy (2006) as an organisation reporting to the city council, comprising of representatives from city agencies and other stakeholders. The UMTA should comprehensively supervise the provision and administration of urban transport, assist in the formulation of sectoral strategy, participate in regulatory activities, transport demand management, and monitor tasks delegated to the implementing organisations. The UMTA has varying degrees of powers, depending on the extent of devolution by the respective state government, in relation to its influence on financing of bus services in cities and providing an enabling environment for establishing a sustainable urban transport system. These powers include issuing licenses and/or permits for offering public transport services, enabling commercialisation of property³, setting performance standards for monitoring bus agencies, and advising on tariffs, fees and charges related to urban transport. Additional powers include facilitation of rationalisation of fare structures and Central or state subsidies, monitoring and auditing the use of Urban Transport Fund (UTF), and funding urban transport. The UTF has multiple direct and indirect fund-collection mechanisms which the UMTA supervises and manages, including budgetary grants, advertising taxes, betterment levies, property taxes and fuel cess.

In many cities, the Gross Cost Contract (GCC) model has been preferred for provision of city bus services, where responsibilities of service planning, fare collection and performance management lie with the private operator under the guidance of UMTA. For example, the Bengaluru UMTA Act (or BMLTA Act, 2022) allocates responsibilities related to issuing licenses and/or permits to public transport services, setting performance standards for monitoring bus agencies and advising on tariffs, fees and charges related to urban transport, monitoring and auditing the use of UTF for city bus services, and urban transport financing.

Uttar Pradesh Dedicated Urban Transport Fund

The multiplicity of responsibilities pertaining to urban transport in India creates uncertainty and confusion relating to the tasks each agency is mandated to perform, which ultimately leads to inefficiencies within agencies and wastage of resources. Considering such resource wastage in the face of limited availability of funds, the provision of integrated transport system infrastructure becomes a major challenge.

Under the JnNURM scheme of Government of India, stimulus was provided to cities having populations more than ten lakhs, for bus rapid transport system or organized urban transport mechanism. In adherence to the directives issued by Ministry of Urban Development for 'Funding for purchase of buses for Urban Transport System under JnNURM', the Uttar Pradesh Government established the Dedicated Urban Transport Fund (DUTF) at the state as well as the city level in 2016. The fund is managed by the

³ Permitting development of commercial activities alongside transport, as deemed appropriate to the context of the property (for instance, a commercial shopping complex atop a bus station)

Uttar Pradesh Dedicated Urban Transport Fund Management Committee and is chaired by the state's Chief Secretary of Urban Development.

The fund is supported by various sources including levy of cess on petrol and diesel, increasing renewal fee for driving licenses, registration of both two and four- wheelers and levying of a congestion fee, fuel cess, green tax, parking fee, property development tax and employment tax (UPUTD, 2016) (UPUTD, The possible sources of Dedicated Urban Transport Fund at city level, 2016). The Union Ministry of Urban Development had also mandated state governments in cities with a population of more than ten lakh to impose a cess on fuel and diesel in order to enhance funds allocated to transport infrastructure.

Electric Vehicle (EV) policies for e-buses

Demand incentives help in propelling early market growth of electric cars, given the country's nascent EV (Electric Vehicle) industry. These might be capital or operational incentives, with the former helping to offset the higher initial costs of EVs and the latter promoting the use of EVs on public roads. Transport and Urban Development Authorities are largely responsible for implementing and enforcing demand incentives at the local and state levels. A suitable combination of demand incentives has been detailed in the state-level EV policies to encourage EV adoption in respective areas. The implementation of road-tax exemptions by state transport agencies complemented by the FAME-II purchase subsidies facilitate faster deployment of EVs. Many states in India have developed EV policies with defined targets for procurement of e-buses and provisions for capital subsidy as well as tax exemption to incentivise procurement of electric buses.

3. Table of insights

Table 1: Financing framework for urban transport and EV policies for E-buses:

 Geography	 Data point – No. of e-buses procured under FAME scheme ⁴	 Data point – % of population using bus as a mode of travel to workplace ⁵	Financing framework for urban transport					Electric Vehicle (EV) policies for e-buses		
			 Proposed establishment of urban transport fund for bus services					 Financial incentives under the state's EV policy for e-buses		
			Source	Provision of Urban Transport Fund	Source of UTF			Source	Capital subsidy provisions for e-buses	Tax exemptions for e-buses
					Land based fiscal tools & development related charges	Cess on other existing taxes and fees	Provision of Operational subsidy for funding support			
Central level	5,595	11.22%	MoHUA - Operations Document for Urban Transport Fund(UTF)	✓ (pg. 19)	✓ (pg. 19)	✓ (pg. 20)	✓ (pg. 14)	Scheme for Faster Adoption and manufacturing of Electric Vehicles in India	✓	✓
Telangana	325	11.21%	Hyderabad Metropolitan Transport Act, 2008	✓ (pg. 46)	✓ (pg. 47)	✓ (pg. 47)	✓ (pg. 48)	Telangana EV policy		✓ (pg. 10)
Maharashtra	725	10.42%	City Mobility Plan for PCMC	✓ (pg. 11)	✓ (pg. 11)	✓ (pg. 11)	✓ (pg. 11)	Maharashtra EV policy	✓ (pg. 21)	✓ (pg. 23)
Karnataka	350	19.13%	Bengaluru Metropolitan Land Transport	✓ (pg. 28)	✓ (pg. 29)	✓ (pg. 29)	✓ (pg. 30)	Karnataka EV Policy		✓ (pg. 31)

⁴: (Ministry of Heavy Industries and Public Enterprise, n.d.)

⁵ (ORGI, 2011)

 Geography	 Data point – No. of e-buses procured under FAME scheme ⁴	 Data point – % of population using bus as a mode of travel to workplace ⁵	Financing framework for urban transport					Electric Vehicle (EV) policies for e-buses		
			 Proposed establishment of urban transport fund for bus services					 Financial incentives under the state's EV policy for e-buses		
			Source	Provision of Urban Transport Fund	Source of UTF			Source	Capital subsidy provisions for e-buses	Tax exemptions for e-buses
					Land based fiscal tools & development related charges	Cess on other existing taxes and fees	Provision of Operational subsidy for funding support			
			Authority Bill, 2021							
Gujarat	550	5.58%	Financing options for urban transport	✓ (pg. 16)		✓ (pg. 17)		Gujarat EV Policy		
			Gujarat VGF Scheme Review by WRI				✓ (pg. 2)			
Delhi	300	21.71%	Delhi EV policy	✓ (pg. 3)		✓ (pg. 3)	✓ (pg. 3)	Delhi EV Policy	No	✓ (pg. 3)
Kerala	250	28.35%	Kerala UMTA Act	✓ (pg. 37)		✓ (pg. 37)	✓ (pg. 37)	Kerala EV Policy	No	✓ (pg. 6)
Tamil Nadu	525	23.13%	-					Tamil Nadu EV Policy	✓ (pg. 17)	No
Rajasthan	100	10.83%	Jaipur City Transport Services Ltd(JCTSL) - Brief Introduction	✓ (pg. 1)			✓ (pg. 5)	Rajasthan EV Policy	✓ (Pg. 9-10)	✓ (Pg. 9)
			Urban Mobility in		✓ (Pg. 5)	✓ (Pg. 5)				

 Geography	 Data point – No. of e-buses procured under FAME scheme ⁴	 Data point – % of population using bus as a mode of travel to workplace ⁵	Financing framework for urban transport					Electric Vehicle (EV) policies for e-buses		
			 Proposed establishment of urban transport fund for bus services					 Financial incentives under the state's EV policy for e-buses		
			Source	Provision of Urban Transport Fund	Source of UTF			Source	Capital subsidy provisions for e-buses	Tax exemptions for e-buses
					Land based fiscal tools & development related charges	Cess on other existing taxes and fees	Provision of Operational subsidy for funding support			
			India Conference							
Haryana	50	9.24%	Gurugram Metropolitan Development Authority Act (GMDA) Act	✓ (pg. 12)			✓ (pg. 13)	Haryana EV Policy	✓ (Pg. 4 and 5)	✓ (Pg. 6)
			NCRTC TOD Policy		✓ (Pg. 10)	✓ (pg. 10)				
Uttar Pradesh	600	3.60%	Uttar Pradesh Urban Transport Directorate	✓ (pg. 1)			✓ (pg. 1)	Uttar Pradesh EV Policy	✓ (Pg. 13)	✓ (Pg. 16)
			Urban Mobility In India Conference, 2016		✓	✓				
Madhya Pradesh	340	5.71%	-					Madhya Pradesh EV Policy	✓ (Pg. 23)	✓ (Pg. 14)
Andhra Pradesh	300	11.21%	-					Andhra Pradesh EV Policy	✓ (Pg. 2)	✓ (Pg. 16)

 Geography	 Data point – No. of e-buses procured under FAME scheme ⁴	 Data point – % of population using bus as a mode of travel to workplace ⁵	Financing framework for urban transport					Electric Vehicle (EV) policies for e-buses		
			 Proposed establishment of urban transport fund for bus services					 Financial incentives under the state's EV policy for e-buses		
			Source	Provision of Urban Transport Fund	Source of UTF			Source	Capital subsidy provisions for e-buses	Tax exemptions for e-buses
					Land based fiscal tools & development related charges	Cess on other existing taxes and fees	Provision of Operational subsidy for funding support			
Chhattisgarh	50	3.39%	Chhattisgarh State EV Policy	✓ (Pg. 24)		✓ (Pg. 24)	✓ (Pg. 15)	Chhattisgarh State EV Policy	✓(pg. 15)	✓ (pg. 16)
Goa		26.48%	Goa Electric Mobility Promotion Policy	✓(pg. 14)		✓ (pg. 14)		Goa Electric Mobility Promotion Policy		✓(pg. 10)
Himachal Pradesh	100	23.89%	Himachal Pradesh EV Policy	✓(pg. 31)		✓(pg. 31)		Himachal Pradesh EV Policy		✓ (pg. 28)
Jharkhand		3.72%						Jharkhand Electric Vehicle Policy	✓(pg. 9)	✓(pg. 10)
Odisha	50	5.35%	Odisha Urban Infrastructure Development Fund	✓			✓	Odisha Electric Vehicle Policy	✓(pg. 9)	✓(pg. 9)
			Odisha Electric Vehicle Policy			✓(pg. 5)				
Punjab		6.37%	The Punjab Urban Transport Fund Act	✓ (pg. 2)		✓ (pg. 8)	✓ (pg. 3)	Punjab Electric Vehicle Policy		✓ (pg. 11)

 Geography	 Data point – No. of e-buses procured under FAME scheme ⁴	 Data point – % of population using bus as a mode of travel to workplace ⁵	Financing framework for urban transport					Electric Vehicle (EV) policies for e-buses		
			 Proposed establishment of urban transport fund for bus services					 Financial incentives under the state's EV policy for e-buses		
			Source	Provision of Urban Transport Fund	Source of UTF			Source	Capital subsidy provisions for e-buses	Tax exemptions for e-buses
					Land based fiscal tools & development related charges	Cess on other existing taxes and fees	Provision of Operational subsidy for funding support			
Uttarakhand	30	7.60%	Uttarakhand State Urban Transport Fund Rules	✓		✓	✓	Uttarakhand Electric Vehicle Policy		
West Bengal	100	9.27%	West Bengal Transport Development Fund Act	✓		✓	✓ (Pg. 4)	West Bengal EV policy	✓ (Pg. 4)	
Bihar	25	2.84%	-					Bihar Draft EV Policy	✓ (Pg. 2)	✓ (pg. 3)

Table 2: Role of UMTAs in city bus services and urban transport financing

 Geography	Role of UMTA in facilitating urban transport financing					
	 Source	 Role in Issuing licenses or permits public transport services to operating agencies	 Setting up performance standards for bus agencies	 Monitoring and advising on tariff, fees and charges related to urban transport	 Monitoring and auditing use of UTF	 Funding or arranging funding for urban transport
Central level	MoHUA Bill	✓ (pg. 16)	✓ (pg. 14)	✓ (pg. 16)	✓ (pg. 16)	✓ (pg. 14)
Telangana	Hyderabad Metropolitan Transport Act, 2008	✓ (pg. 51)		✓ (pg. 51)	✓ (pg. 49)	✓ (pg. 17)
Maharashtra	MMRDA website					
Karnataka	BENGALURU Metropolitan Land Transport Authority Bill, 2021	✓ (Pg. 14)	✓ (Pg. 14)	✓ (Pg. 14)	✓ (Pg. 19)	✓ (Pg. 14)
Gujarat	-					
	-					
Delhi	-					
Kerala	Kerala UMTA Act	✓ (pg. 37)	✓ (pg. 34)	✓ (pg. 37)	✓ (pg. 37)	✓ (pg. 33)
Tamil Nadu	Chennai UMTA Act			✓ (pg. 3)		✓ (pg. 4)
Rajasthan	-					
	-					
Haryana	Gurugram Metropolitan Development Authority Act (GMDA) Act	✓ (pg. 13)		✓ (pg. 13)	✓ (pg. 17)	✓ (pg. 13)
Uttar Pradesh	-					
Madhya Pradesh	-					
Andhra Pradesh	-					
Chhattisgarh	-					

 Geography	Role of UMTA in facilitating urban transport financing					
	 Source	 Role in Issuing licenses or permits public transport services to operating agencies	 Setting up performance standards for bus agencies	 Monitoring and advising on tariff, fees and charges related to urban transport	 Monitoring and auditing use of UTF	 Funding or arranging funding for urban transport
Goa	-					
Himachal Pradesh	Constitution of 23-member Unified metropolitan transport authority					✓
Jharkhand	-					
Odisha	-					
Punjab	Constitution of 23-member Unified metropolitan transport authority					✓
Uttarakhand	-					
West Bengal	-					
Bihar	-					

4. References

- CEPT University. (2013). *Service Level Benchmark in Urban Transport for Indian Cities*. New Delhi: Ministry of Urban Development. https://mohua.gov.in/upload/uploadfiles/files/VoulmeI_Methodologyreport_final03.pdf
- Delhi Integrated Multi-Modal Transit Systems. (2016). *Roadmap for improving City Bus Systems in India*. New Delhi, India: Shakti Sustainable Energy Foundation. <https://smartnet.niua.org/sites/default/files/resources/Roadmap%20for%20improving%20City%20Bus%20Systems%20in%20India.pdf>
- Deodhar, R. (2017, October 31). *Improving BEST: Revenue Generation of Mumbai's public transporter – Part 2*. MoneyLife: <https://www.moneylife.in/article/improving-best-revenue-generation-of-mumbais-public-transporter-part2/52034.html>
- Hook, W., & Hughes, C., et al. (2019). *Best Practice in National Support for Urban Transportation Part 2: Growing Rapid Transit Infrastructure – Funding, Financing, and Capacity*. Institute for Transportation and Development Policy (ITDP). <https://itdpdotorg.wpengine.com/wp-content/uploads/2015/11/Best-Practice-in-National-Support-for-Urban-Transport-Part-2-Funding-Finance-and-Capacity-1.pdf>
- Ministry of Housing and Urban Affairs. (2021, April). *Completion Report: Efficient and Sustainable City Bus Services (ESCBS) project*. <https://mohua.gov.in/upload/uploadfiles/files/Completion%20Report%20-ESCBS%20Project.pdf>
- Ministry of Heavy Industries and Public Enterprise. (n.d.). *Sanction of electric buses under Phase-II of Faster Adoption and Manufacturing of Electric Vehicles in India Scheme*. New Delhi: Ministry of Heavy Industries. https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&ved=2ahUKEwjL_qOzv8WDAxVgxigGHVU3D-EQFnoECBoQAQ&url=https%3A%2F%2Fframe2.heavyindustries.gov.in%2FWriteReadData%2Fuserfiles%2FPress%2520Release%2520for%2520Buses.pdf&usg=AOvVaw3WPvD_cqK4T4on9KN
- ORGI. (2011, n.d. n.d.). *Data visualizations: Mode of commute for workers*. Census of India: <https://censusindia.gov.in/census.website/>
- UPUTD. (2016). *The possible sources of Dedicated Urban Transport Fund at city level*. Lucknow: Uttar Pradesh Urban Transport Directorate. Retrieved from https://uputd.gov.in/site/writereaddata/uploadedCirculars/C_201612231505474600.pdf
- UPUTD. (2016). *The possible sources of Dedicated Urban Transport Fund at State level*. Lucknow: Uttar Pradesh Urban Transport Directorate. Retrieved from https://uputd.gov.in/site/writereaddata/uploadedCirculars/C_201612231505012363.pdf
- World Resources Institute (WRI) India. (2021, May). *National investment program for bus based public transport systems in India*. Shakti Sustainable Energy Foundation. <https://shaktifoundation.in/wp-content/uploads/2021/11/Annex-1-National-Investment-program-for-bus-based-public-transport-systems-in-India.pdf>



NITI Aayog

www.niti.gov.in